

Graph-CoRe: Graph Representation Learning with Contrastive Subgraph Replacement

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Abstract—The interpretability of Graph Neural Networks (GNNs) is crucial for enhancing trustworthiness and practical utility in various domains like finance, healthcare, and transportation. Given the unique challenges posed by graph data, traditional interpretability methods fall short in deciphering the intricate workings of GNNs. In this study, we propose a subgraph replacement strategy combined with supervised contrastive learning to improve the interpretability and predictive capabilities of GNNs when processing graph data. By introducing perturbations during training and addressing data biases, our approach enhances model robustness, stability, and key feature identification. Moreover, by mitigating label imbalance issues through tailored data augmentation techniques, we ensure a more balanced and accurate prediction model. Experimental results demonstrate the effectiveness of our proposed method in outperforming other subgraph representation techniques across diverse datasets, showcasing its utility in enhancing model interpretability and performance.

Index Terms—subgraph replacement, interpretability, graph representation learning, supervised contrastive learning

I. INTRODUCTION

The interpretability of Graph Neural Networks (GNNs) pertains to their capability to furnish clear and rational explanations for the model's forecasts and reasoning paths. In numerous practical scenarios, such as financial risk management, medical diagnostics, and intelligent transport systems, it is imperative for the model's decision-making process to be transparent and understandable, enabling users to trust and heed the model's advice [1], [2]. Particularly in contexts involving sensitive or high-stakes decisions, the absence of interpretability can culminate in errors and biased outcomes, leading to significant social and ethical repercussions [3]. Hence, examining the interpretability of GNNs is vital for boosting their trustworthiness and practical utility [4]. Interpretable GNNs facilitate user comprehension of the decision-making

rationale, assessment of model reliability and precision, and provide insights for model refinement and enhancement.

Given its sparsity and non-Euclidean nature, graph data differs significantly from Euclidean data, impeding the application of conventional interpretability methods. The multifaceted architecture of GNNs, encompassing numerous neural network layers and graph convolution operations, renders traceability and interpretation of model determinations challenging [5]. Comprehending these intricate computational procedures demands advanced knowledge and high-degree abstraction aptitude. As such, tailored methodologies are required to elucidate GNN behavior when processing graph data. In real-world settings, graph data often carries noise and uncertainty. Clarifying how GNNs manage this inherent volatility, pinpointing pivotal features tightly linked to labels, and safeguarding the model's robustness and reliability constitute crucial challenges.

Interpreting Graph Neural Networks through subgraph replacement is one of the key research approaches in this field. Drawing on invariant learning principles [6]–[8], this study employs a subgraph replacement strategy to facilitate the model's identification of crucial graph features. By integrating subgraph replacement perturbations into the training distribution, we generate various augmented datasets, enabling the model to discern stable key feature-label relationships across these datasets and thus improve generalization to new data. Such perturbations during training address data biases, boost model robustness and stability in practical applications, simplify key factor identification, enhance model interpretability, and deepen user insights into the model's workings and predictions. Especially in situations of data scarcity, the subgraph replacement technique is vital for training deep learning models, as larger datasets generally enhance model generalizability and performance. While the DIR method [9] partitions

the graph into factor and non-factor components from a causal analysis standpoint, it struggles to represent the factor component in small datasets effectively. Conversely, the GERA method [10] categorizes the graph into essential subgraphs and environmental subgraphs through subgraph replacement data augmentation, yet overlooks potential imbalance issues introduced by the augmentation. The subgraph replacement approach proficiently creates new samples, expanding the dataset's size. Yet, considering the original data's label imbalance, augmentation techniques could worsen this disparity. Hence, this research mitigates such imbalance by adjusting the proportions of labels within the augmented data.

This study creates a model named Graph-CoRe by integrating subgraph replacement data augmentation techniques, class balance methods, and the supervised contrastive learning approach, aiming to obtain superior representations of key subgraphs. The supervised contrastive learning approach extends from self-supervised to supervised contrastive learning, fully leveraging label information and data augmentation. In the embedding space, the model clusters samples of the same category and their augmented versions together while distancing samples of different categories, thus forming tighter clusters. This process facilitates the model in learning effective representations more efficiently.

Our contributions can be summarized as follows:

- The primary achievements comprise the fusion of subgraph replacement with contrastive learning, which bolsters key subgraph identification and feature learning capabilities, and enhances the predictive interpretability of graph neural networks.
- By employing label balancing techniques, the complications introduced by data augmentation are mitigated, consequently reducing the model's prediction bias towards certain categories.
- Experimental outcomes demonstrate that the graph classification model Graph-CoRe proposed herein, surpasses other key subgraph representation methods across a wide array of datasets.

II. RELATED WORK

A. Interpretability of Graph Neural Networks

Several methods exist for explainable Graph Neural Networks (GNNs), and we encapsulate pertinent studies from a perturbation perspective here. Achieving interpretability of GNNs via perturbation chiefly involves generating masks in varied ways to highlight key input features. Depending on the explanation task, different masks, like node, edge, or node feature masks, are produced. These masks are then fused with the input graph to form a new graph encapsulating significant input information. Subsequently, this graph is fed into a trained GNN to assess the masks and modify the mask generation algorithm. The critical input features flagged by the masks should impart substantial semantic meanings, and utilizing these captured features for predictions should yield results akin to those from the original graph [5]. GNNExplainer

[11] pioneered the application of soft masks for this purpose, optimizing the masks by maximizing the mutual information between the predictions of the original and masked graphs. However, the soft attribute of these masks introduces noise-related challenges. GREA [10] adopts soft masks to segregate molecules into key and non-key parts within the latent vectors of hidden layers which simplifies graph segmentation and enhances efficiencies. By combining vectors of both partitions, new samples are generated. However, this process is prone to data label imbalance. AD-GCL [12] employs a Bernoulli distribution to generate masks, deciding which graph edges to remove for enhanced graph representation. Through adversarial training, it optimizes the mutual information between the original and augmented graphs while minimizing extracted information, enabling the encoder to capture the minimal necessary information for each graph's identification. Both PGExplainer [13] and GraphMask [14] concentrate on edge masking to preserve the integrity of the graph's predictive capacity while discarding redundant information. Specifically, PGExplainer employs a reparameterization technique to address the discretization issue of masks, intending to mitigate the noise typically associated with soft masks. ZORRO [15] applies a greedy algorithm to yield discrete masks for critical input nodes and features, bypassing the non-differentiability challenge of discrete masks and reducing potential inaccuracies due to noise. Yet, the greedy mask selection might lead to locally optimal explanations, and as separate masks are created for each graph, the explanations can lack a global perspective. SubgraphX [16] investigates subgraph-centric explanations, employing a Monte Carlo tree search to identify and rationalize influential subgraphs, contributing to a nuanced understanding of model predictions. DIR [9] adopts a causal perspective, using soft masks to distinguish between causative and non-causative graph constituents, aiming to uncover invariant causal relationships and improve GNN interpretability through intervention-oriented methodologies. Causal Screening [17] focuses on the causal attribution of edges in graph predictions, employing Individual Causal Effect (ICE) measures to discern the impact of specific edges, thus refining the explanation process through a causal lens. ReFine [18] proposes a multi-granular explanation framework, capitalizing on the predictive mutual information between the original and perturbed graphs to learn discrete mask predictors, providing insights from both global and local perspectives.

B. Contrastive Learning Methods

Graph Neural Networks (GNNs) represent a broad field within graph representation learning. Here, we primarily focus on topics most relevant to graph contrastive learning (GCL). Initially proposed for feature extraction through data augmentation and leveraging large networks, contrastive learning was later applied to convolutional neural networks (CNNs) for image representation learning [19]–[24]. GCL extends the principles of contrastive learning (CL) to GNNs. However, compared to CNNs, employing GNNs in GCL introduces new challenges. In images, contrastive learning typically generates

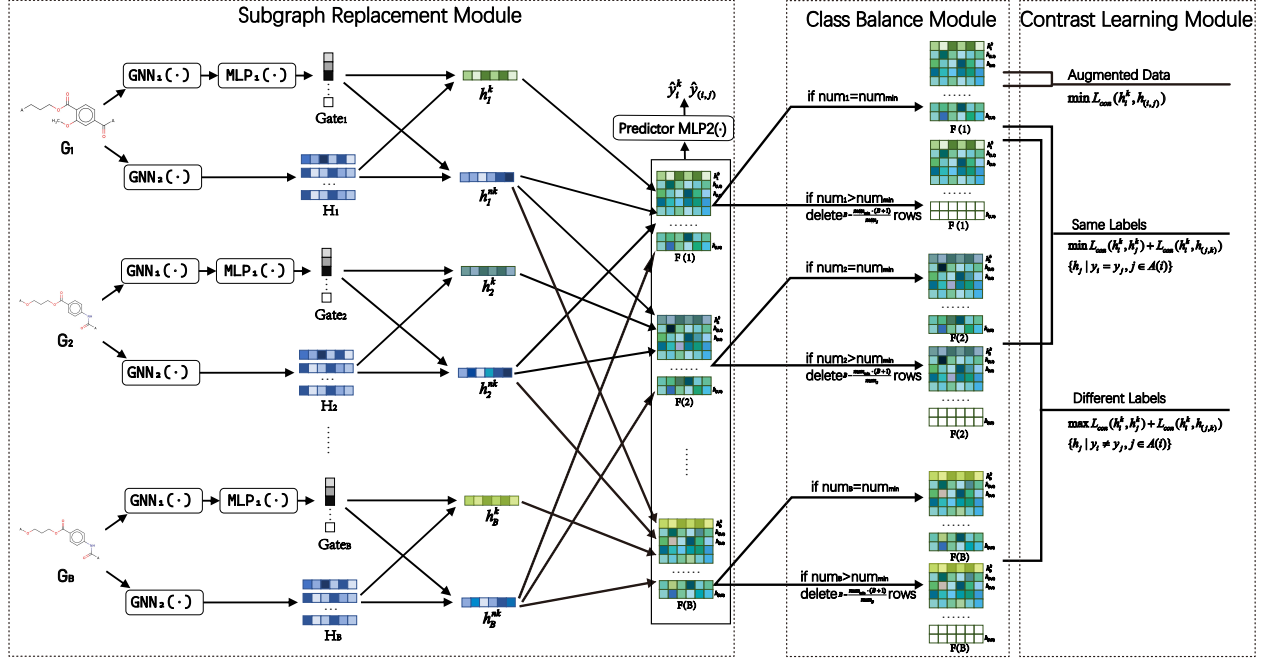


Fig. 1. The overview of Graph-CoRe. Subgraph Replacement Module: This step encodes the graph to determine the soft mask *Gate* and the corresponding node representation *H* for each graph. These elements combine to yield the key and non-key subgraph representation vectors. Label Balancing Module: New augmented data is obtained by combining the resulting vector representations. The augmented data is then trimmed according to the number of label categories. Contrastive Learning Module: Positive and negative samples are identified based on label consistency and augmentation status, followed by the implementation of supervised contrastive learning

positive samples through data augmentation techniques such as rotation, scaling, cropping, etc. In contrast, due to the irregularity and abstraction of graphs, designing natural contrastive pairs becomes more complex. Therefore, designing contrastive pairs for GCL, which must deal with irregular graph structures, presents additional challenges. Common methods include random node deletion, edge perturbation, subgraph sampling, and feature masking. Recently, GraphCL [25] introduced four GDA operations: random node deletion (along with its edges), random addition or deletion of edges, random masking of node attributes, and sampling of subgraphs. These operations create many different graph views for contrastive learning. MVGRL [26] employs diffusion graphs as a second view for contrastive learning. According to their research, two views (the original graph and a diffusion graph) perform better than three views (the original graph, a diffusion graph, and a heat kernel graph). JOAO [27] uses an adaptive and automated solution to learn augmentation strategies, modeling the selection of GDA as a two-level optimization problem with the outer level learning the augmentation strategy and the inner level learning the graph representation given the augmentation. AD-GCL [12] combines adversarial training and contrastive learning to maximize mutual information between the original and enhanced graphs, while minimizing the extracted information. This allows the encoder to capture only the minimal information necessary for each graph identification,

leading to improved graph representation. The DGI model [28] performs contrastive learning by maximizing the mutual information between the global and local representations of a graph. The InfoGCL model [29] takes into account the amount of information in a graph, applying data augmentation methods and dealing with key information in the graph structure and node features when generating contrastive pairs.

III. OUR PROPOSED METHOD

This paper presents a graph contrastive learning approach premised on subgraph replacement, aimed at enhancing the interpretability and prediction precision of Graph Neural Networks (GNNs). This method boosts the discernment of crucial graph components via subgraph replacement, category balancing, and contrastive learning. It includes four modules: a subgraph replacement module, a category balancing module, a contrastive learning module, and a prediction module. The overall framework of the proposed method is illustrated in Figure 1.

A. Subgraph Replacement

This study uses a perturbation strategy based on subgraph replacement to effectively demarcate key and non-key sections within the graph data. This generates multiple intervention distributions enabling the model to learn stable relationships between key features and labels across various

distributions, thereby enhancing its generalization to new data. Inspired by the GREA method [10], this paper partitions graphs in latent space, significantly reducing computational complexity, avoiding the intricacy of explicit graph decoding and encoding, hence increasing the model's efficiency. The encoder $GNN_1(\cdot)$ is employed to encode the input graph data G_i , thus deriving the representation of each node $H_i = [h_1, h_2, \dots, h_n]^T \in R^{n \times d}$ within G_i . This is further mapped via the encoder $MLP_1(\cdot)$ and activation function σ , ultimately producing the gate vector $Gate_i \in R^{n \times d}$, which matches the dimensionality of the node representation. $Gate_i$ controls the selection of each node's information in G_i , allowing certain node information to pass through while blocking the transmission of others. Additionally, it quantifies the likelihood of classifying each node into a key subgraph, while $1_N - Gate_i$ represents the probability of allocating a node to a non-key subgraph. Splitting G_i using $Gate_i$ yields the key subgraph g_i^k and non-key subgraph g_i^{nk} .

$$Gate_i = \sigma(MLP_1(GNN_1(G_i))) \quad (1)$$

Encode the input graph data G_i using another encoder $GNN_2(\cdot)$ to obtain representations for each node $H_i = [h_1, h_2, \dots, h_n]^T \in R^{n \times d}$.

$$H_i = [h_1, h_2, \dots, h_n]^T \in R^{n \times d} = GNN_2(G_i) \in R^{n \times d} \quad (2)$$

With $Gate$ and H , the key subgraph and non-key subgraph can be easily separated in the latent space. Using sum pooling, we have,

$$\begin{aligned} h_i^k &= \text{pooling}(1_N^T \cdot (Gate_i \times H_i)) \\ h_i^{nk} &= \text{pooling}(1_N^T \cdot ((1_N - Gate_i) \times H_i)) \end{aligned} \quad (3)$$

Where 1_N denotes the N-size column vector with all entries as 1, and h_i^k and h_i^{nk} are the representation vectors of key subgraph and non-key subgraph, respectively. The objective of dividing the graph into key and non-key subgraphs is to isolate key subgraph representation vectors that eliminate interference and noise, ensuring these vectors encapsulate all necessary features for prediction. Consequently, it suffices to utilize these key subgraph representation vectors for accurately predicting the graph's properties. That is, given the key subgraph representation h_i^k of graph G_i , the predicted label is,

$$\hat{y}_i^k = MLP_2(h_i^k) \quad (4)$$

Non-key subgraphs pertain to natural-state noise or biases. For enhancing model robustness against noise, this paper integrates the key subgraph representation vectors with non-key vectors from other graphs, yielding several synthetic graph representation vectors. This method aids in providing more accurate graph representation vectors. As the key subgraphs incorporate various non-key subgraph interferences, the model can learn more correspondences between key subgraphs and labels, enabling it to identify more representative subgraph structures and learn more effective key subgraph representation vectors. During training, with each batch size being B, it is

possible to obtain B sets of subgraph representation vectors $((h_1^k, h_1^{nk}), (h_2^k, h_2^{nk}), \dots, (h_B^k, h_B^{nk}))$. For each graph G_i , we amalgamate its key subgraph vector h_i^k not only with its concurrent non-key subgraph vector h_i^{nk} , but also with all other non-subgraph vectors h_j^{nk} , where $j \in \{1, 2, \dots, B\} \setminus \{i\}$ within the batch. The integration of the key subgraph representation vector h_i^k and non-key subgraph representation vector h_j^{nk} is achieved via an aggregation function $AGG(\cdot)$. Applicable aggregation functions encompass various combining/pooling tactics such as concatenation, sum pooling, and max pooling. For the obtained synthetic graph expression vector $h_{i,j}$, the predicted label $\hat{y}_{i,j}$ is mapped by the predictor $MLP_2(\cdot)$.

$$\begin{aligned} h_{(i,j)} &= AGG(h_i^k, h_j^{nk}) \\ \hat{y}_{(i,j)} &= MLP_2(h_{(i,j)}) \end{aligned} \quad (5)$$

The key subgraph representation vectors h_i^k and synthetic graph representation vectors $h_{(i,j)}$ of graph G_i are forecasted using the consistent predictor $MLP_2(\cdot)$, both sharing the true labels y_i corresponding to graph G_i . The individual loss function values of the key subgraph and synthetic graph representation vectors for graph G_i are respectively determined using the cross-entropy loss function.

$$\begin{aligned} L_k &= - \sum_{c=1}^M y_{i,c} \log(\hat{y}_{i,c}^k) \\ L_{com} &= \frac{1}{B} \sum_{j=1}^B L_j = - \frac{1}{B} \sum_{j=1}^B \sum_{c=1}^M y_{i,c} \log(\hat{y}_{(i,j),c}) \end{aligned} \quad (6)$$

Here, L_k denotes the loss function for the key subgraph, while L_{com} refers to the loss function for the synthetic graph, derived from this key subgraph and other non-key subgraphs in the batch. The i th class is represented by c , and M designates the total number of classes. $y_{i,c}$ is a binary indicator function, returning 1 if the graph's true class is equal to c , otherwise returning 0. The synthetic graph's predicted probability for class c is expressed as $\hat{y}_{(i,j),c}$. To manage the extent of total graph information allocated by the key subgraphs, this study introduces a regularization term as specified below,

$$L_{reg} = \left| \frac{1_N^T \cdot Gate_i}{N} - \gamma \right| \quad (7)$$

Where γ is a hyperparameter, the expected ratio of information in key subgraphs to the total graph, and when deviating from this value, it is penalized.

B. Supervised Contrastive Learning for Data Augmentation

Leveraging the subgraph replacement method, this approach yields multiple augmented datasets, with the contrastive learning method allowing finer control over the retained information in the representation vector [30], [31]. Contrastive learning aims to differentiate by comparing sample similarities and distinctions, aiding in the development of significant representations and the extraction of critical data features. It

narrows the gap between an anchor and a positive sample in the embedding space while distancing the anchor from numerous negative samples. In unsupervised learning, without labels, positive samples often derive from augmentations, and negatives from random mini-batch selections. This could waste valuable label information and risk poor representation by potentially selecting negative samples with the same label [32].

The SupCon method [19] employs label information for supervised learning, enhancing model accuracy and robustness by clustering together samples and their augmentations from the same class in the embedding space and separating those from different classes. For a graph G_i , its key subgraph expression vector h_i^k and any synthetic graph expression vector $h_{(i,j)}$ serve as augmented samples with identical true labels to G_i . Thus, for graph G_i as the anchor, the B augmented samples including h_i^k and $B - 1$ synthetic vectors $h_{(i,j)}$ act as positive samples, while samples sharing the anchor's label are also positive, and those with differing labels are negative.

$$L_{con} = \sum_{i \in I} \mathcal{L}_{con,i} = \sum_{i \in I} \frac{-1}{|H_p|} \sum_{h_p \in H_p} \log \frac{\exp(h_i \cdot h_p / t)}{\sum_{h_a \in D(i)} \exp(h_i \cdot h_a / t)} \quad (8)$$

Here, $\cdot$ signifies the inner product of vectors, i is marked as the anchor, and $t \in R^+$ symbolizes a scalar temperature parameter. I includes all graph data samples, excluding their augmented samples. $A(i) = I \setminus \{i\}$ describes all graph data, excluding the anchor node. The set $F(i) \equiv \{h_i^k, h_{(i,j)}\}$ includes the expression vectors of augmented samples tied to graphs, incorporating key subgraph expression vectors of graph G_i and their synthetic graph vectors. $H_p \equiv F(i) \cup \{h_j \cup F(j) | y_i = y_j, j \in A(i)\}$ represents all positive sample expression vectors, inclusive of each graph G_i 's key subgraphs and its corresponding synthetic graph vectors, as well as the expression vectors of key subgraphs and synthetic graphs from other graph datasets with a label matching the main graph G_i . The notation $|H_p|$ denotes the quantity within set H_p . $D(i) \equiv F(i) \cup \{h_j \cup F(j) | j \in A(i)\}$ designates all expression vectors without including the graph G_i . This incorporates the key subgraphs and synthetic graphs' expression vectors for graph G_i , other graph data, and the corresponding key subgraphs and synthetic graph vectors for these data.

C. Class Balancing

Due to label imbalance in the dataset, subgraph replacement significantly multiplies the label count per class, intensifying the imbalance issue. If some classes vastly outnumber others in the training set, the model might favor these classes, weakening its predictive capability for lesser-represented classes. This skew can drastically impact model performance, particularly when test data categories differ from those in training. Consequently, enhancements to address label imbalance are imperative when employing this data augmentation technique.

As discussed in the previous sections, the number of augmented samples derived from each graph is B , i.e., the quantity increases from one to $B + 1$ graphs as depicted in Figure 2, where $B = 64$ is showcased using PROTEINS,

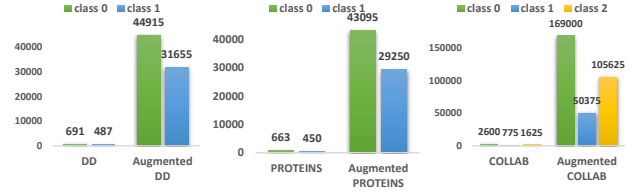


Fig. 2. PROTEINS, DD, COLLAB dataset Class Difference Visualization before and after data augmentation

DD, and COLLAB datasets. The parameter B could be chosen from $\{32, 64, 128, 256\}$. However, post-augmentation, the class difference in the graph data significantly amplifies. Owing to label imbalance, it's vital to dynamically adjust the number of augmented samples for each graph G_i . In any batch, the number of each label is represented as $(num_1, num_2, \dots, num_c, \dots, num_M)$. Here, $num_{\min}$ refers to the label with the least instances, for which the number of augmented samples is set at B . Other classes are adjusted relative to this benchmark. For instance, if the current label count is num_c , its augmented samples total to $\frac{num_{\min} \cdot (B+1)}{num_c}$.

D. Model Training Optimization

During training, the model is bifurcated into two components: the prediction model and the segmentation model. These models are iteratively optimized, with the loss function of the prediction model and the segmentation model defined as in Equations (9).

$$\begin{aligned} L_{pred} &= L_k + \alpha L_{com} + \beta L_{con} \\ L_{sep} &= L_k + \alpha L_{com} + \beta L_{con} + \lambda L_{reg} \end{aligned} \quad (9)$$

Here, α, β, λ is a tunable parameter employed to regulate the weight of each component within the loss function. $L_k, L_{com}, L_{con}, L_{reg}$ represent the loss functions of the key subgraph, synthetic graph, contrastive learning, and regularization respectively. Notably, the contrastive learning loss function is the aggregate loss function of the key subgraph and synthetic graph.

IV. EXPERIMENTS

In this section, we assess the performance of models grounded in Graph-CoRe.

A. Dataset and Baseline

1) *Datasets*: This study examines the efficacy of graph contrastive learning models for subgraph replacement, utilizing two primary dataset types: biochemical-related and social network datasets. The biochemical datasets implement anticancer activity classification, protein structure analysis, and toxicology prediction, represented by NCI1 [33], DD [34], PROTEIN [35], and PTC [36] respectively. The social network datasets, including COLLAB [37], IMDB-BINARY [37], and Reddit [38], explore collaborative relationships among researchers, actors, and Reddit users. Refer to TABLE II for detailed dataset characteristics.

TABLE I
GRAPH CLASSIFICATION PERFORMANCE COMPARISON

	NCI	PROTEINS	DD	PTC_MR	COLLAB	REDDIT-BINARY	REDDIT-MULTI-5K	IMDB-BINARY
WL	80.01±0.50	72.92±0.56	74.02±2.28	58.0±0.50	69.30±3.44	68.82±0.41	46.06±0.21	72.30±3.44
DGK	80.31±0.46	73.30±0.82	74.85±0.74	60.10±2.60	64.66±0.50	78.04±0.39	41.27±0.18	66.96±0.56
node2vec	54.89±1.61	57.49±3.57	-	58.60±8.00	56.10±0.20	-	-	-
sub2vec	52.84±1.47	53.03±5.55	54.33±2.44	60.00±6.40	55.26±1.54	71.48±0.41	36.68±0.42	55.26±1.54
graph2vec	73.22±1.81	73.30±2.05	70.32±2.32	60.20±6.90	71.10±0.54	75.78±1.03	47.86±0.26	71.10±0.54
MVGRL	77.00±0.80	-	-	62.50±1.70	76.00±1.20	84.50±0.60	-	74.20±0.70
InfoGraph	76.20±1.06	74.44±0.31	72.85±1.78	61.70±1.40	70.65±1.13	82.50±1.42	53.46±1.03	73.03±0.87
GraphCL	77.87±0.41	74.39±0.45	72.62±0.40	61.30±2.10	71.36±1.15	89.53±0.84	55.99±0.28	71.14±0.44
JOAO	78.07±0.47	74.55±0.41	77.32±0.54	-	69.50±0.36	85.29±1.35	55.74±0.63	70.21±3.08
JOAOv2	78.36±0.53	74.07±1.10	77.40±1.15	-	69.33±0.34	89.42±1.45	56.03±0.27	71.83±0.25
SimGRACE	79.12±0.44	75.35±0.09	77.44±1.11	-	71.72±0.82	89.51±0.89	55.91±0.34	71.30±0.77
EG-SimCLR	82.97±0.20	75.44±0.65	78.86±0.51	61.51±2.41	76.60±1.26	90.70±0.46	56.22±0.56	71.78±0.34
EG-Simsiam	81.49±0.19	76.06±0.51	78.35±0.83	62.67±1.50	74.73±0.79	89.17±0.43	56.37±0.17	71.93±0.40
GREa	89.34±1.02	73.47±2.87	81.84±1.85	64.55±6.70	91.38±0.35	96.10±0.87	84.73±0.14	82.28±1.07
GIN+GIB	74.3 ± 3.3	74.9 ± 5.1	74.7 ± 3.9	55.8 ± 2.0	78.5 ± 1.6	-	-	73.7 ± 7.0
KERGNN-2	76.3±2.6	75.5±4.6	78.9±3.5	56.38±3.38	72.7±2.1	82.0±2.5	48.35±12.38	73.7±4.0
Graph-CoRe	90.09±0.65	75.84±0.80	82.27±1.77	70.07±1.83	92.44±0.17	96.63±0.18	85.12±0.29	82.94±0.53

TABLE II
STATISTICS AND PROPERTIES OF THE DATASET

Datasets	graph numbers	graph classes	average nodes	average edges
NCI	4110	2	29.8	32.3
PROTEIN	1113	2	39.06	72.82
DD	1178	2	284.32	715.66
PTC_MR	344	2	14.29	14.69
COLLAB	5000	3	74.49	2457.78
REDDIT-BINARY	2000	2	429.63	497.75
REDDIT-MULTI-5K	4999	5	508.52	594.87
IMDB-BINARY	1000	2	19.77	96.53

2) *Baseline*: This chapter contrasts Graph-CoRe with related approaches to elucidate its performance in graph classification. The methods are categorized as follows:

- Graph Kernel Methods for Graph Classification: These methods focus on graph representation learning, notable examples include WL [39], DGK [37]. They utilize graph kernels to capture graph similarities.
- Graph Contrastive Learning Methods to Enhance Graph Classification Performance: This category includes techniques like GraphCL [40], JOAO [41], JOAO(v2) [41], SimGRACE [42] and MVGRL [43], which leverage contrastive learning principles to improve graph classification outcomes.
- Interpretable Graph Classification Methods: Focused on pinpointing critical information within graph data, examples here include GREa [10], GIB [44], KERGNN [45]. These methods aim to enhance the interpretability of graph classifications.

The comparison leverages the area under the ROC curve (AUC) as the evaluative metric for graph classification tasks, presenting mean and standard deviations from quintuple experiments to ensure reliability. GIN [46] is uniformly applied as the encoder across all experiments to maintain experimental fairness.

B. Experimental Results

1) *Effectiveness*: Graph-CoRe significantly outperforms both graph kernel methods and graph contrastive learning methods across all datasets. Compared to interpretable graph neural network methods, Graph-CoRe shows a slight 0.22% performance degradation on the PROTEINS dataset against the EG-Simsiam method but demonstrates substantial improvements on all other datasets. The exceptional performance of Graph-CoRe can be attributed to its deeper feature extraction capabilities. During the segmentation of graph data, Graph-CoRe considers semantic issues, avoiding the destruction of critical structures. Using learnable masks, the graph data is divided into key subgraphs and non-key subgraphs, and combining them generates synthetic graphs that increase the diversity of the graph data. Supervised graph contrastive learning methods are more accurate in the selection of positive and negative samples. Such strategies help the model to learn the representations of key subgraphs more deeply and treat non-key subgraphs as interference, thus training a more robust model.

Compared to the method GREa, which is similar to Graph-CoRe, Graph-CoRe achieved performance improvements on all datasets, especially on the PTC_MR dataset, where the performance increase was 5.52%, surpassing the improvements seen on other datasets. This is due to the smaller sample size of the PTC_MR dataset, where Graph-CoRe effectively utilized the synthetic augmented data through its graph contrastive learning module, further learning valuable information and significantly improving the prediction accuracy of the graph classification model on smaller datasets.

2) *Ablation Experiments*: This study uses the PROTEINS and DD datasets as exemplars to investigate the class balance module's impact on model performance through an ablation experiment. After removing the class balance module, a significant decrease in the average AUC and a noticeable upward trend in the variance of AUC for both datasets are observed, as demonstrated in TABLE III. This implies the class

TABLE III
THE AUC COMPARISON RESULTS OF THE REMOVAL BALANCE MODULE

	PROTEINS	DD
Graph-CoRe	75.84±0.80	82.27±1.77
Graph-CoRe-no balance	75.36±1.58	79.67±2.02

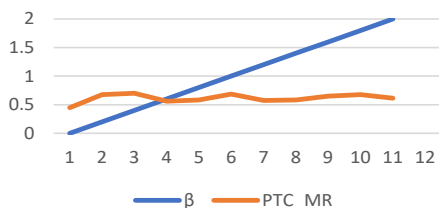


Fig. 3. Results of Sensitivity Analysis for Hyper-parameters β of the Dataset PTC_MR

balance module’s crucial role in mitigating class imbalance implications and enhancing model stability.

Further analysis reveals that removing the balancing mechanism results in the model’s underperformance when handling samples from minority classes, impacting the overall predictability and dependability. Thus, designing and integrating the class balance module is essential to improve the model’s classification capability and generalizability. These experimental results provide a valuable reference for further studies on class balance mechanisms, leading to potential enhancements in the model structure and training strategy. Ultimately, such improvements aim to better accommodate unbalanced datasets in real-world applications, thereby improving the model’s practical effectiveness.

Parameter sensitivity analysis: In the experiment based on the PTC_MR dataset, with all parameters fixed except for β , we evaluate the sensitivity of the parameter β to the experimental prediction result AUC. As can be seen from Figure 3, β has some effects on AUC but does not show a clear pattern.”

3) *Explainability Analysis*: The Gate value, within a range of $[0,1]$, governs how much content each node should retain, employing a soft mask method that simplifies gradient calculation. An interpretability analysis is conducted using the NCI1 dataset. The dataset does not specify the atoms represented by each node’s label; we use consistent numbers and colors for the same atom. Nodes with a Gate value exceeding 0.5 are treated as key subgraphs. Figure 4 illustrates two graphs from the NCI1 dataset with labels 0 and 1; the left graph is label 0 and the right graph is label 1. Key subgraph nodes are marked by red boxes. The observation results indicate that the key subgraphs extracted by the Graph-CoRe model are relatively continuous and can form functional groups with certain chemical properties. This finding suggests that the Graph-CoRe model can effectively extract key subgraphs (functional groups) from the graph, thereby predicting the properties of biochemical molecules. This reveals the intrinsic mechanism of the model’s predictions and further validates the effectiveness of the Graph-CoRe model in the task of

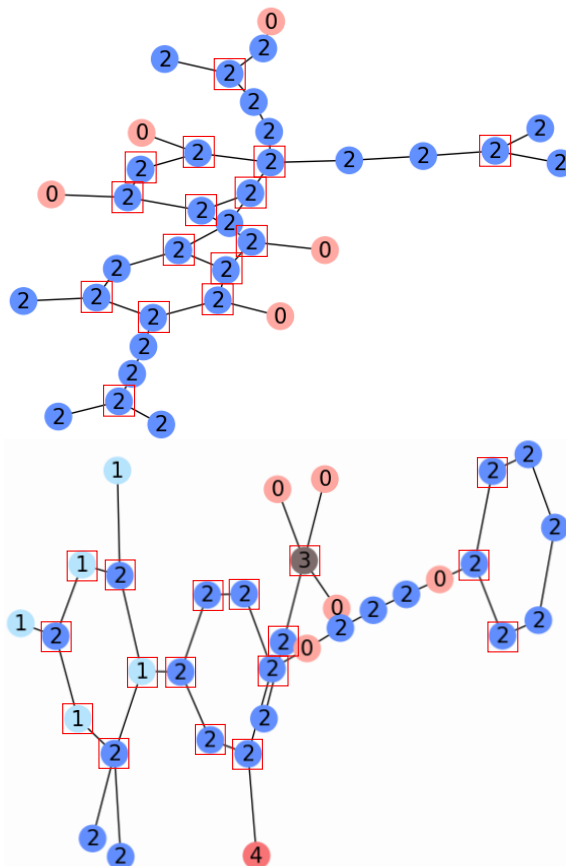


Fig. 4. Key subgraphs in NCI1 dataset display

predicting the properties of biochemical molecules.

V. CONCLUSION

We propose an effective framework that segments real data into key subgraphs and non-key subgraphs in a latent space, creating enhanced data through combination, and conducting supervised contrastive learning with both real and augmented data. The effectiveness of this approach is proven through experiments on both biochemical and social network datasets. The method only judges the effectiveness of the extracted key subgraphs through AUC and visualization methods. In the future, it is necessary to design more accurate evaluation metrics for the interpretability of graph neural networks, which can be used to assess the interpretability of the model.

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